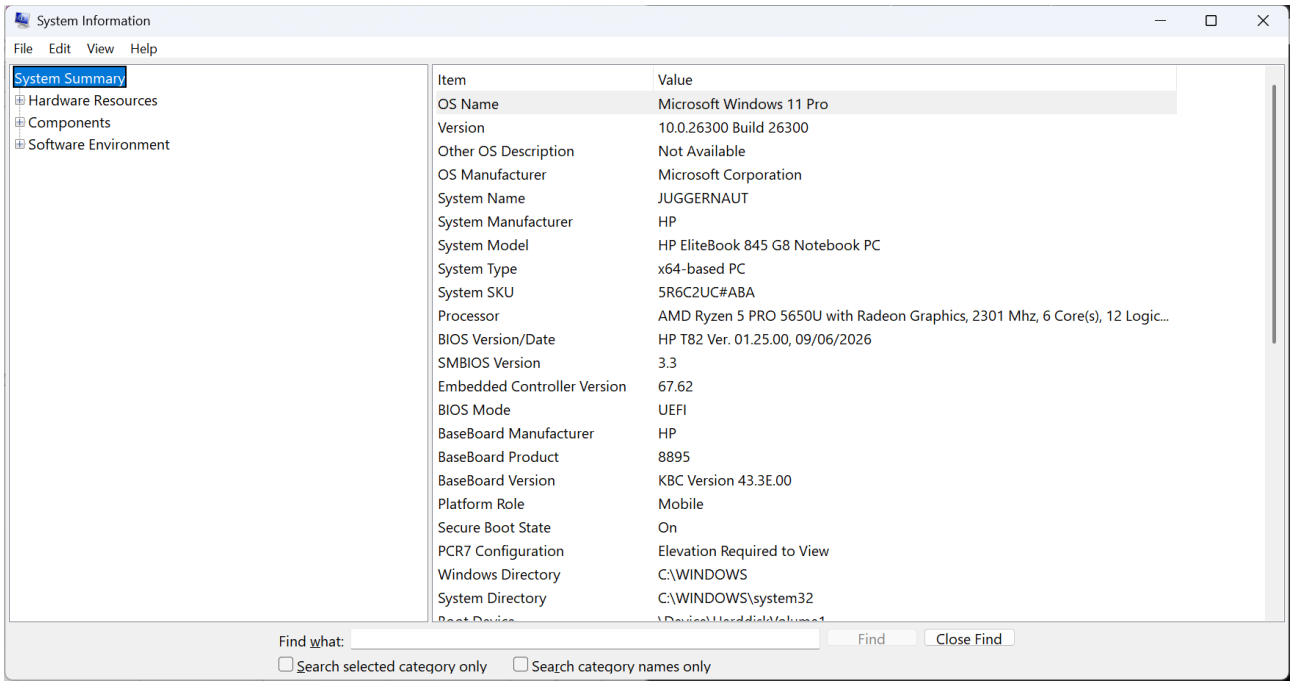
Full name: Adeleke Oluwapelumi Faruq

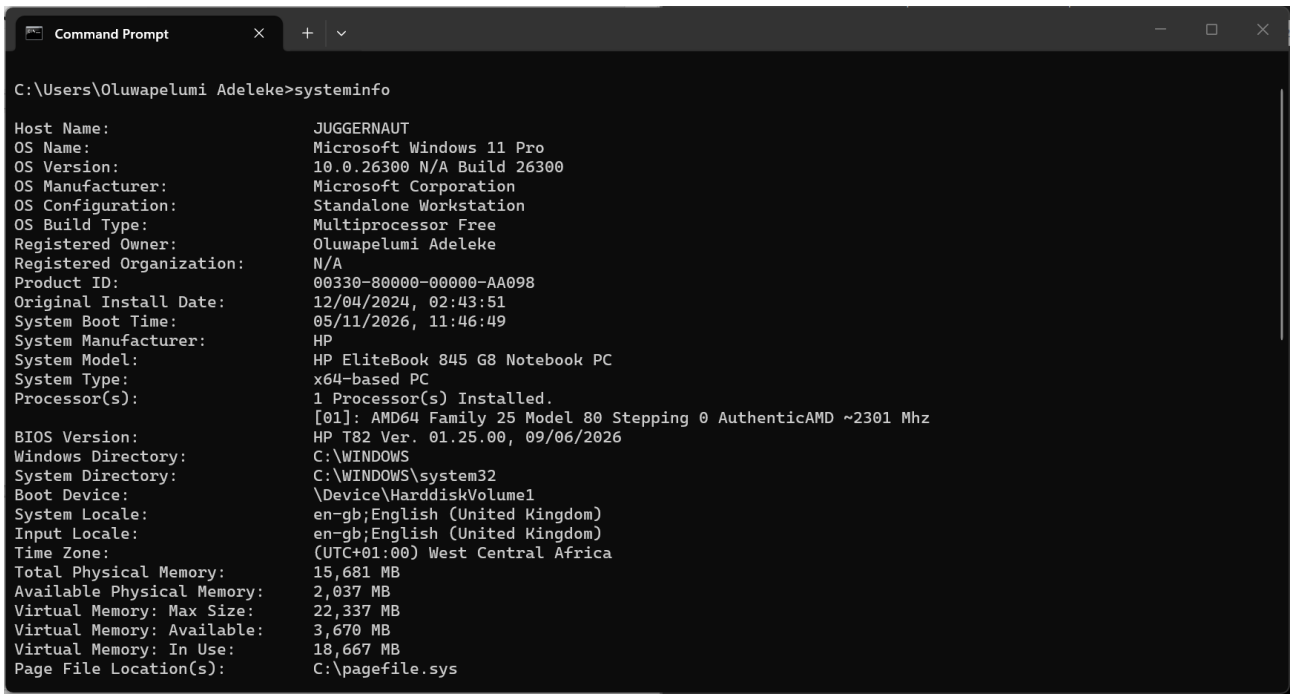
Registration number: C11/26/FCDF/17167

Linux distribution / version: Kali GNU/ 2025.1

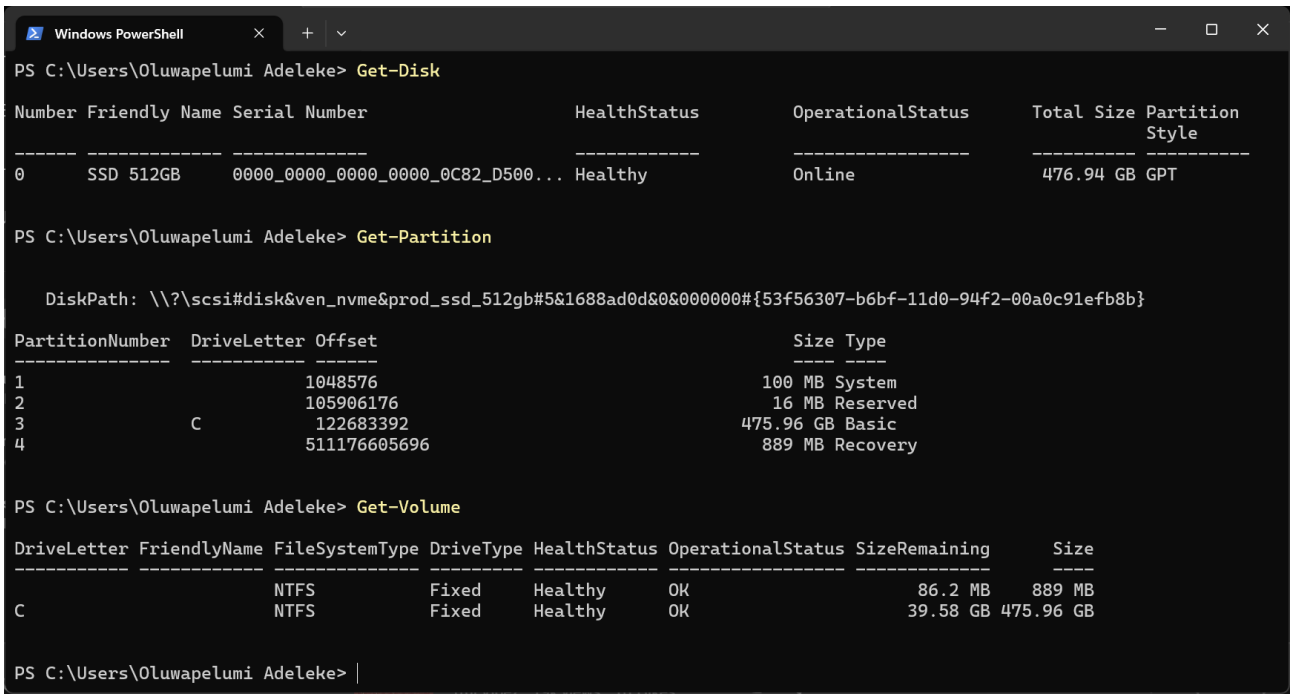
VM hostname: Oracle VirtualBox

A1





A2

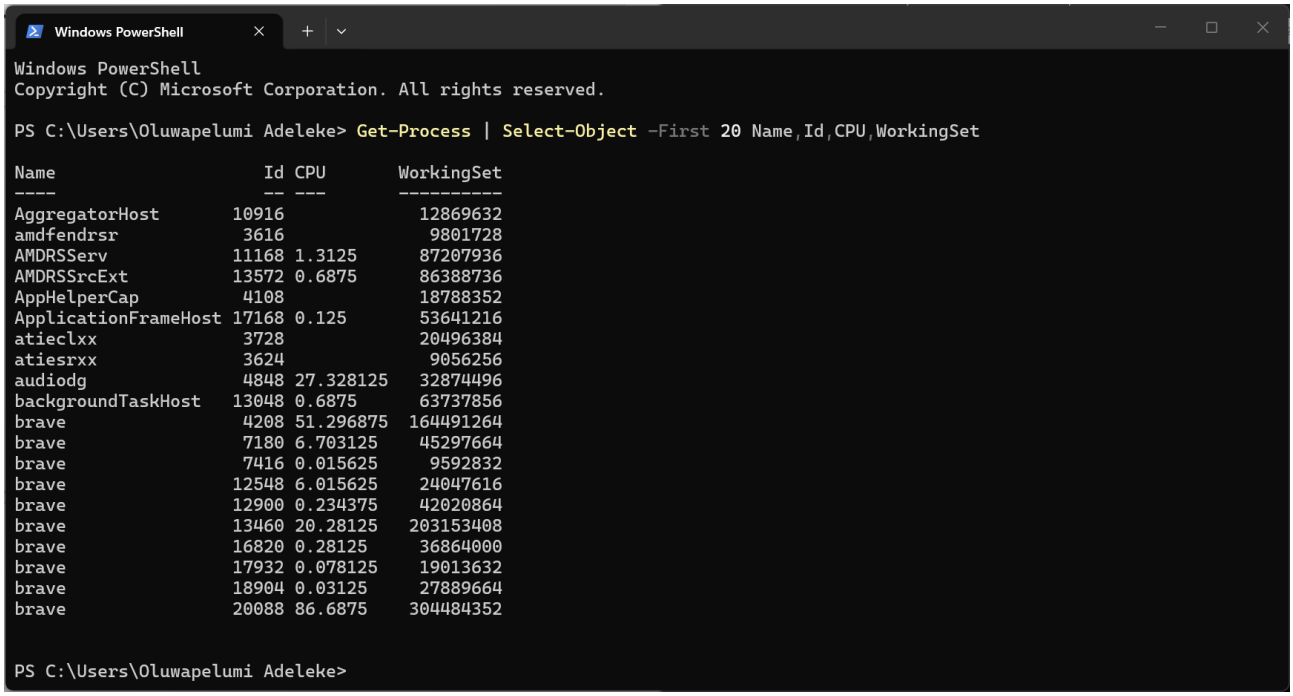


A3

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Typed search queries, recently opened documents, command-line execution history, saved credentials, and desktop arrangement.	UserAssist, RecentDocs, TypedPaths, or PowerShell	Directly attributes specific actions, intent, and knowledge to an individual user account logged into the workstation at a specific timestamp.
Application layer	Web browsing history, downloaded files, cached images, email databases, chat logs, and Office document author metadata.	Application folders and databases	Reconstructs high level user behaviour, external communications, data exfiltration vectors, and the specific software utility used to create or modify contraband files.
OS layer	Account authentication records, privilege escalations, USB device connection histories, background service installations, and application execution counts.	Windows logs, registry, Linux /var/log	Establishes an authoritative system timeline, proves whether external storage media was attached, tracks lateral movement, and verifies program execution even if the application was later uninstalled.
Disk layer	Deleted files, file system metadata records, file slack fragments, unallocated cluster remnants, and hidden volume partitions.	Master File Table ($MFT), $UsnJrnl, Inode tables, Partition Tables (MBR Sector 0 / GPT LBA 1 –33), Volume Slack, and Unallocated Disk Space.	Enables data carving and recovery of intentionally deleted evidence, reveals anti-forensic timestomping (by comparing $STANDARD_INFORMATION vs. $FILE_NAME timestamps), and exposes hidden partitions.
Memory/process layer	Active processes, injected DLLs/shellcode, plaintext cryptographic keys, active TCP/UDP network sockets, clipboard contents, and fileless malware.	RAM capture, process list	Captures live system state and transient artifacts that never touch the hard drive, providing the only way to detect memory-only malware, active C2 connections, and unencrypted disk keys.
Network & Hardware/Firmware Layer	MAC addresses, ARP caches, DNS resolver caches, Wi-Fi SSID connection profiles, UEFI boot variables, and PCI/USB hardware IDs.	RAM network structures, NVRAM/CMOS UEFI variables, setupapi.dev.log	Correlates the physical machine with network router logs, proves physical proximity via historical Wi-Fi SSIDs, and detects low-level firmware persistence.

Explain why evidence may exist in more than one layer:
Modern computing follows the principle of Cascading Layer Interaction (User Activities = Applications + Operating System + Disk/Memory). A user cannot interact directly with physical storage sectors, every single action traverses the entire technology stack, leaving independent footprints at each level.

B1



B2

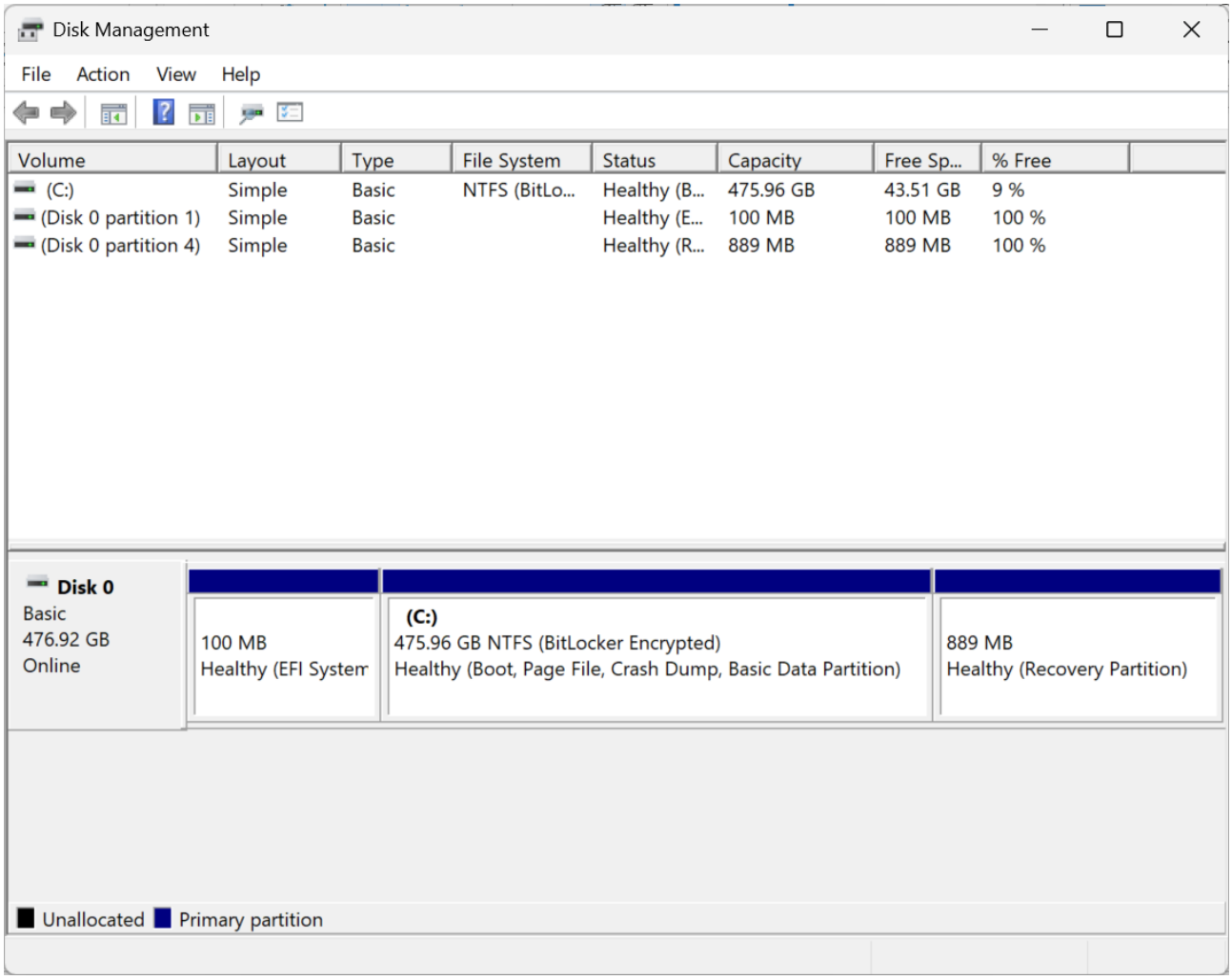
- Program: A set of instructions, code, and data stored as a file on a computer drive.
- Process: An active, running instance of a program that has been loaded into your computer's memory (RAM) and is being managed by the operating system.
- Thread: The smallest unit of execution inside a process.

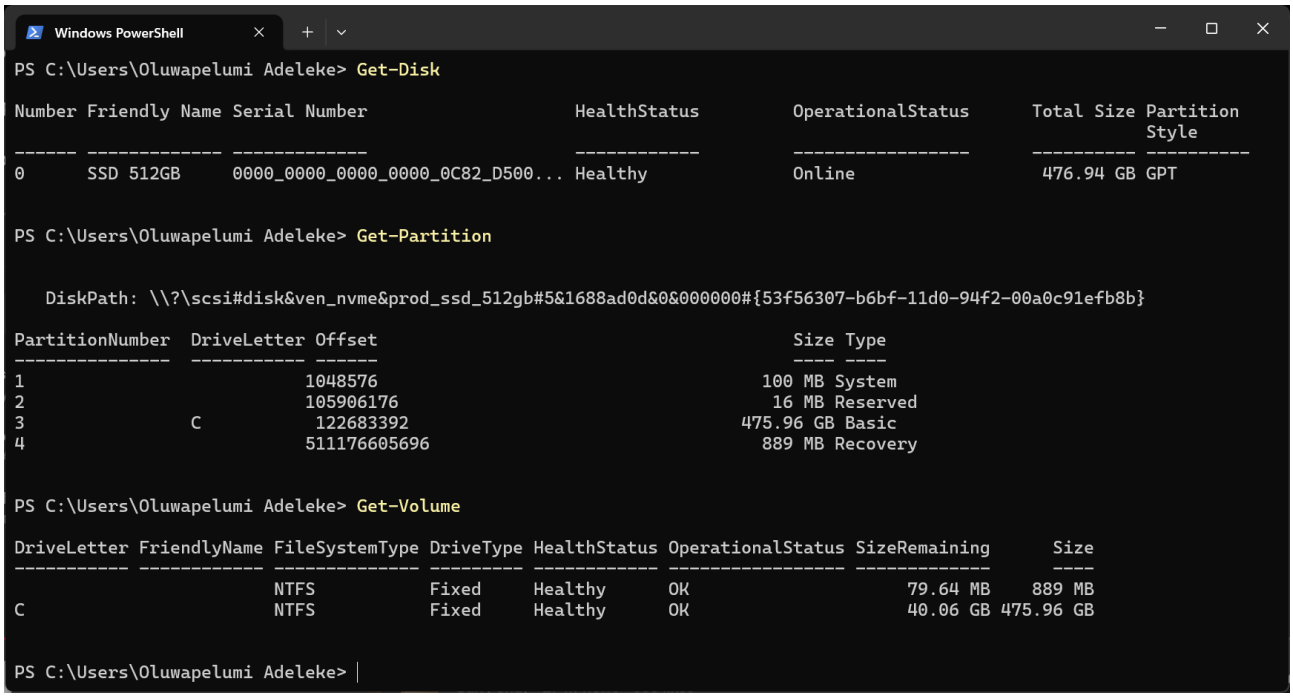
C1.

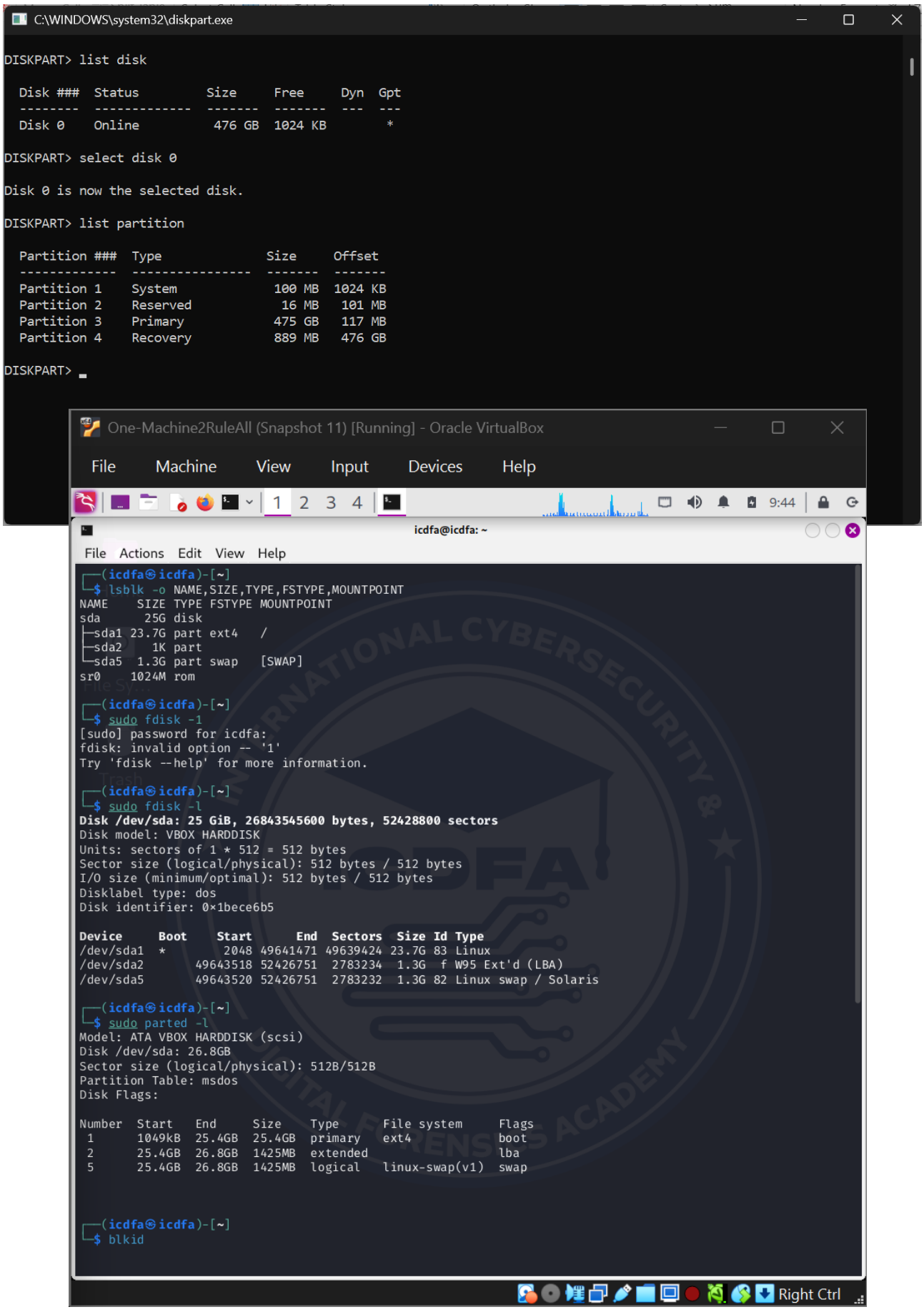
Item	Formula / Value	Answer
Total sectors	204,800	204,800 sectors
Total bytes	204,800 x 512	104,857,600 bytes
Kilobytes	104,857,600 / 1024	102,400 KB
Megabytes	102,400 / 1024	100.00 MB
Gigabytes	100.00 / 1024	0.10 GB

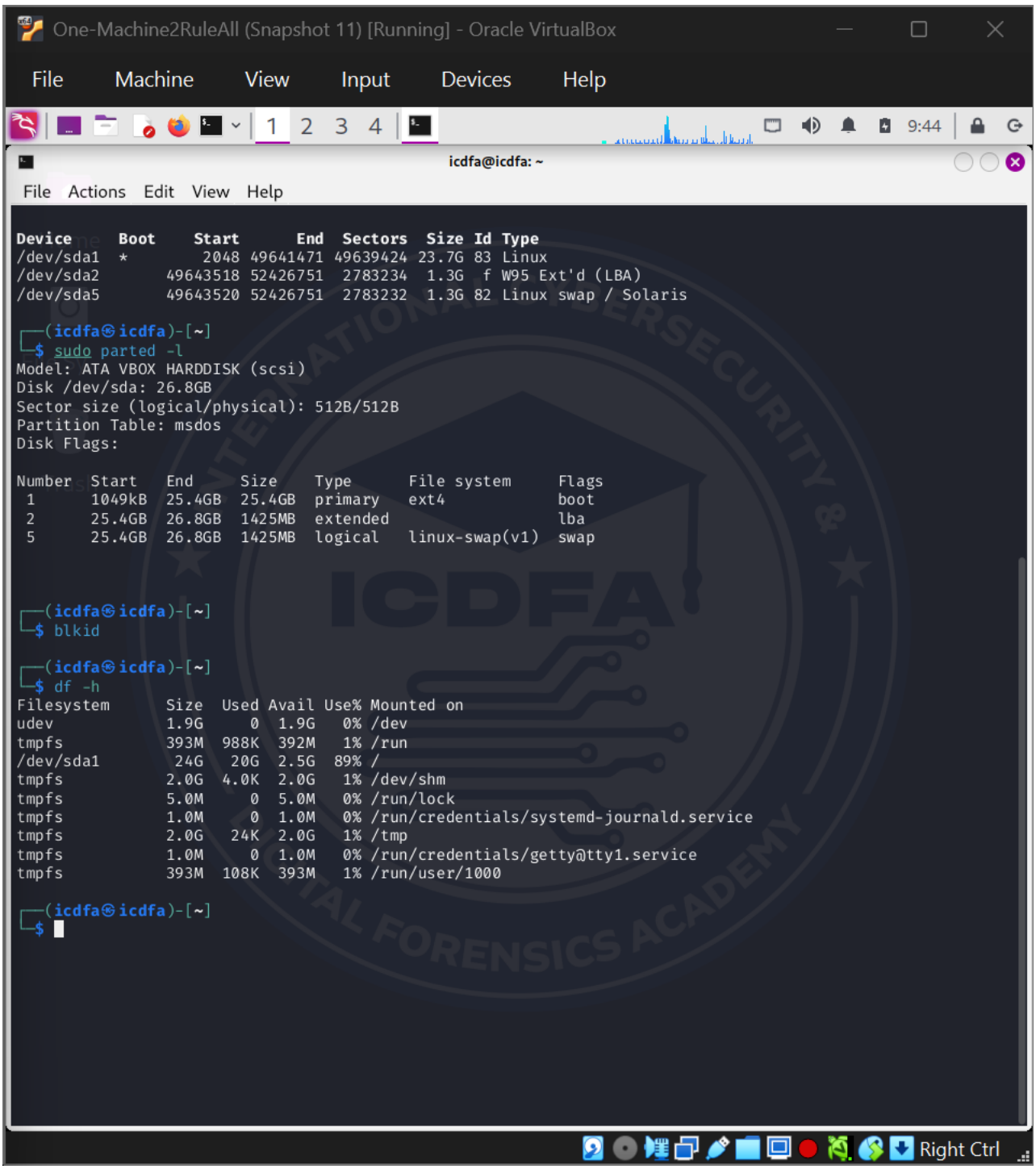
Item	Formula / Value	Answer
Total sectors	32,768	32,768 sectors
Total bytes	32,768 x 512	16,777,216 bytes
Kilobytes	16,777,216 / 1024	16,384 KB
Megabytes	16,384 / 1024	16.00 MB
Gigabytes	16.00/ 1024	0.02 GB

D1


Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
(C:) Simple Basic NTFS (BitLo... Healthy (B... 475.96 GB 43.51 GB 9 %
(Disk 0 partition 1) Simple Basic Healthy (E... 100 MB 100 MB 100 %
(Disk 0 partition 4) Simple Basic Healthy (R... 889 MB 889 MB 100 %
Disk 0
Basic
476.92 GB
Online
100 MB
Healthy (EFI System
(C:)
475.96 GB NTFS (BitLocker Encrypted)
Healthy (Boot, Page File, Crash Dump, Basic Data Partition)
889 MB
Healthy (Recovery Partition)
Unallocated
Primary partition


Windows PowerShell
PS C:\Users\Oluwapelumi Adeleke> Get-Disk
Number Friendly Name Serial Number HealthStatus OperationalStatus Total Size Partition Style
0 SSD 512GB 0000_0000_0000_0000_0C82_D500... Healthy Online 476.94 GB GPT
PS C:\Users\Oluwapelumi Adeleke> Get-Partition
DiskPath: \\?\scsi#disk&ven_nvme&prod_ssd_512gb#5&1688ad0d&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
PartitionNumber DriveLetter Offset Size Type
1 1048576 100 MB System
2 105906176 16 MB Reserved
3 C 122683392 475.96 GB Basic
4 511176605696 889 MB Recovery
PS C:\Users\Oluwapelumi Adeleke> Get-Volume
DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining Size
NTFS Fixed Healthy OK 79.64 MB 889 MB
C NTFS Fixed Healthy OK 40.06 GB 475.96 GB
PS C:\Users\Oluwapelumi Adeleke>


C:\WINDOWS\system32\diskpart.exe
DISKPART> list disk
Disk ### Status Size Free Dyn Gpt
Disk 0 Online 476 GB 1024 KB *
DISKPART> select disk 0
Disk 0 is now the selected disk.
DISKPART> list partition
Partition ### Type Size Offset
Partition 1 System 100 MB 1024 KB
Partition 2 Reserved 16 MB 101 MB
Partition 3 Primary 475 GB 117 MB
Partition 4 Recovery 889 MB 476 GB
DISKPART>
One-Machine2RuleAll (Snapshot 11) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~
File Actions Edit View Help
(icdfa㉿icdfa)-[~]
$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME SIZE TYPE FSTYPE MOUNTPOINT
sda 25G disk
├─sda1 23.7G part ext4 /
├─sda2 1K part
└─sda5 1.3G part swap [SWAP]
sr0 1024M rom
(icdfa㉿icdfa)-[~]
$ sudo fdisk -1
[sudo] password for icdfa:
fdisk: invalid option -- '1'
Try 'fdisk --help' for more information.
(icdfa㉿icdfa)-[~]
$ sudo fdisk -l
Disk /dev/sda: 25 GiB, 26843545600 bytes, 52428800 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1bece6b5
Device Boot Start End Sectors Size Id Type
/dev/sda1 * 2048 49641471 49639424 23.7G 83 Linux
/dev/sda2 49643518 52426751 2783234 1.3G f W95 Ext'd (LBA)
/dev/sda5 49643520 52426751 2783232 1.3G 82 Linux swap / Solaris
(icdfa㉿icdfa)-[~]
$ sudo parted -l
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 26.8GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:
Number Start End Size Type File system Flags
1 1049kB 25.4GB 25.4GB primary ext4 boot
2 25.4GB 26.8GB 1425MB extended lba
5 25.4GB 26.8GB 1425MB logical linux-swap(v1) swap
(icdfa㉿icdfa)-[~]
$ blkid
Right Ctrl



E1

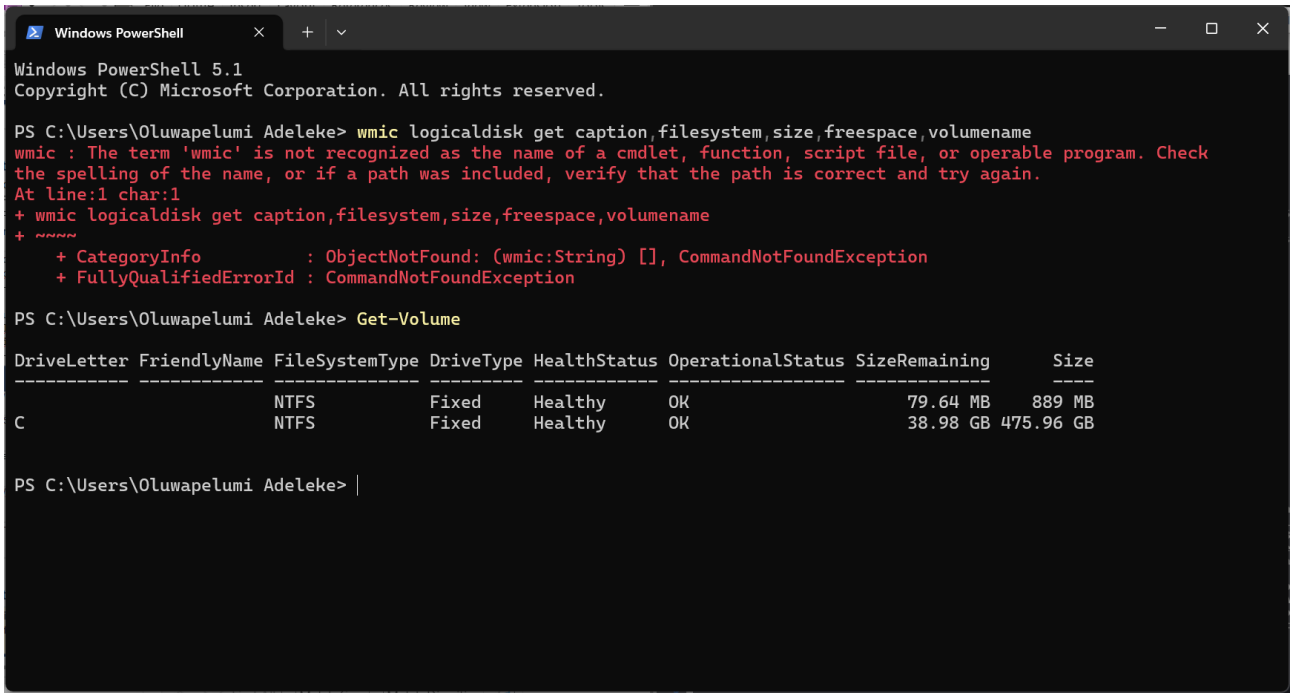
PC BOOT Process

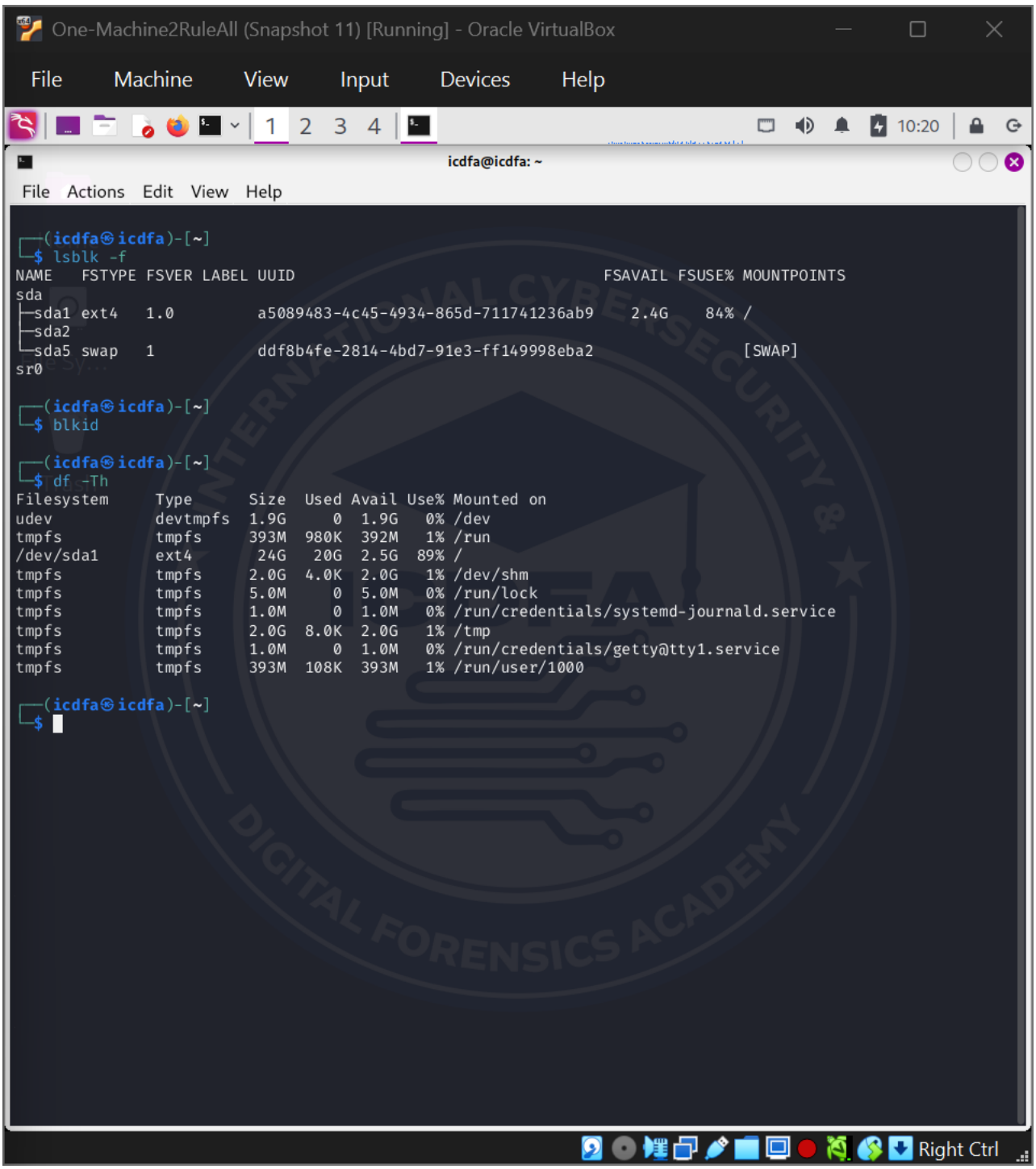
1. **Power-On and Power-On Self-Test (POST)**: Pressing the power button stabilizes electrical current from the PSU and sends a POWER_GOOD signal to the motherboard. The CPU resets its instruction pointer to read firmware in ROM and executes POST, checking that essential hardware (RAM, CPU, storage controllers, display, keyboard) is functional.
2. **BIOS / UEFI Firmware & CMOS/NVRAM Settings:** The motherboard firmware (legacy BIOS or modern UEFI) reads configuration parameters stored in non-volatile CMOS/NVRAM memory including the system real-time clock (RTC), drive controller modes (AHCI/NVMe/RAID), Secure Boot keys, and the boot device priority order.

3. **MBR / GPT or EFI Boot Record Resolution**: In Legacy BIOS/MBR mode: The firmware reads Sector 0 (MBR) of the primary boot drive, executes the 446-byte bootstrap code, parses the 64-byte partition table to find the single partition flagged as Active (0x80), and jumps to that partition's Volume Boot Record (VBR).
4. **Boot Loader / Boot Manager Stage:** The firmware hands control to the OS boot manager. On UEFI Windows systems, \EFI\Microsoft\Boot\bootmgfw.efi executes and reads the Boot Configuration Data (BCD) registry hive to identify installed operating systems. On Linux systems, GRUB2 (grubx64.efi) loads its configuration (grub.cfg).
5. **Kernel Loading and OS Control Initialization:** The boot loader loads the core operating system kernel into RAM (winload.efi loads ntoskrnl.exe, hal.dll, and the SYSTEM Registry hive on Windows; GRUB loads vmlinuz and initramfs on Linux). The kernel initializes protected memory management, loads boot-start device drivers, mounts the root file system, and launches user-space session managers.

Based on the evidence captured in msinfo32, my computer uses UEFI (BIOS Mode: UEFI)

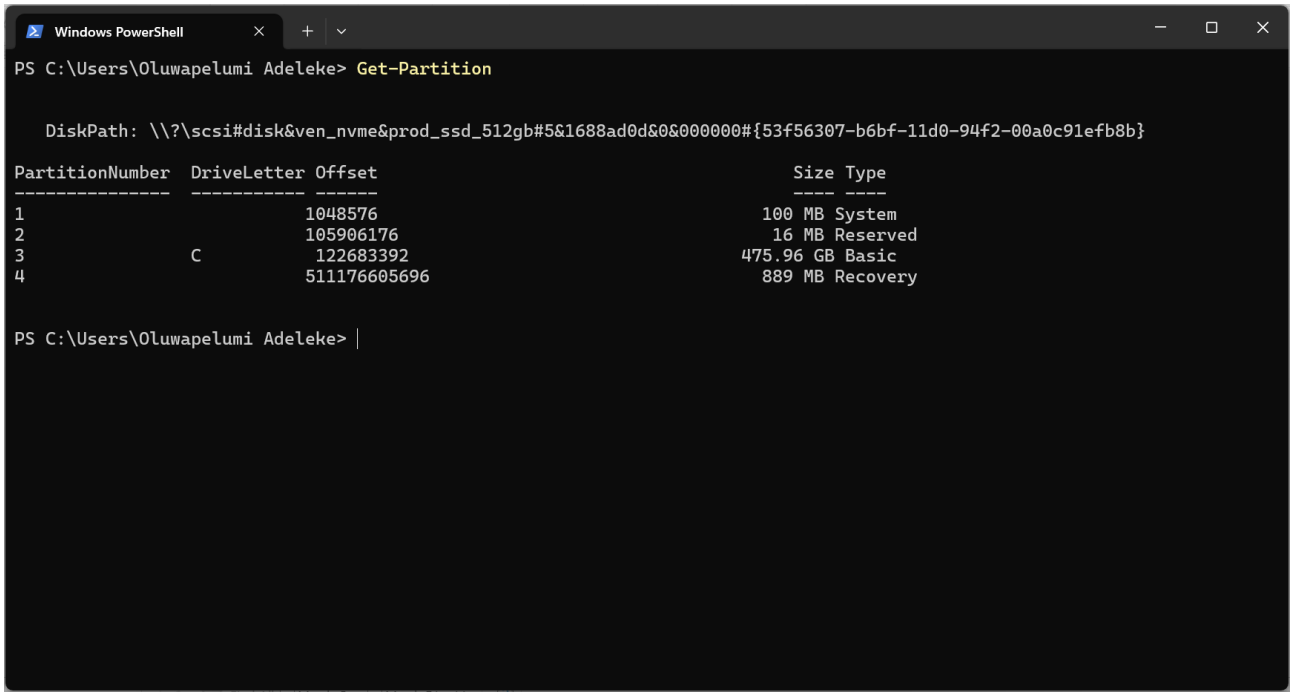
F




One-Machine2RuleAll (Snapshot 11) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
10:20
icdfa@icdfa: ~
File Actions Edit View Help
┌──(icdfa㉿icdfa)-[~]
└─$ lsblk -f
NAME FSTYPE FSVER LABEL UUID FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1 ext4 1.0 a5089483-4c45-4934-865d-711741236ab9 2.4G 84% /
├─sda2
└─sda5 swap 1 ddf8b4fe-2814-4bd7-91e3-ff149998eba2 [SWAP]
sr0
┌──(icdfa㉿icdfa)-[~]
└─$ blkid
┌──(icdfa㉿icdfa)-[~]
└─$ df -Th
Filesystem Type Size Used Avail Use% Mounted on
udev devtmpfs 1.9G 0 1.9G 0% /dev
tmpfs tmpfs 393M 980K 392M 1% /run
/dev/sda1 ext4 24G 20G 2.5G 89% /
tmpfs tmpfs 2.0G 4.0K 2.0G 1% /dev/shm
tmpfs tmpfs 5.0M 0 5.0M 0% /run/lock
tmpfs tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs tmpfs 2.0G 8.0K 2.0G 1% /tmp
tmpfs tmpfs 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service
tmpfs tmpfs 393M 108K 393M 1% /run/user/1000
┌──(icdfa㉿icdfa)-[~]
└─$
Right Ctrl

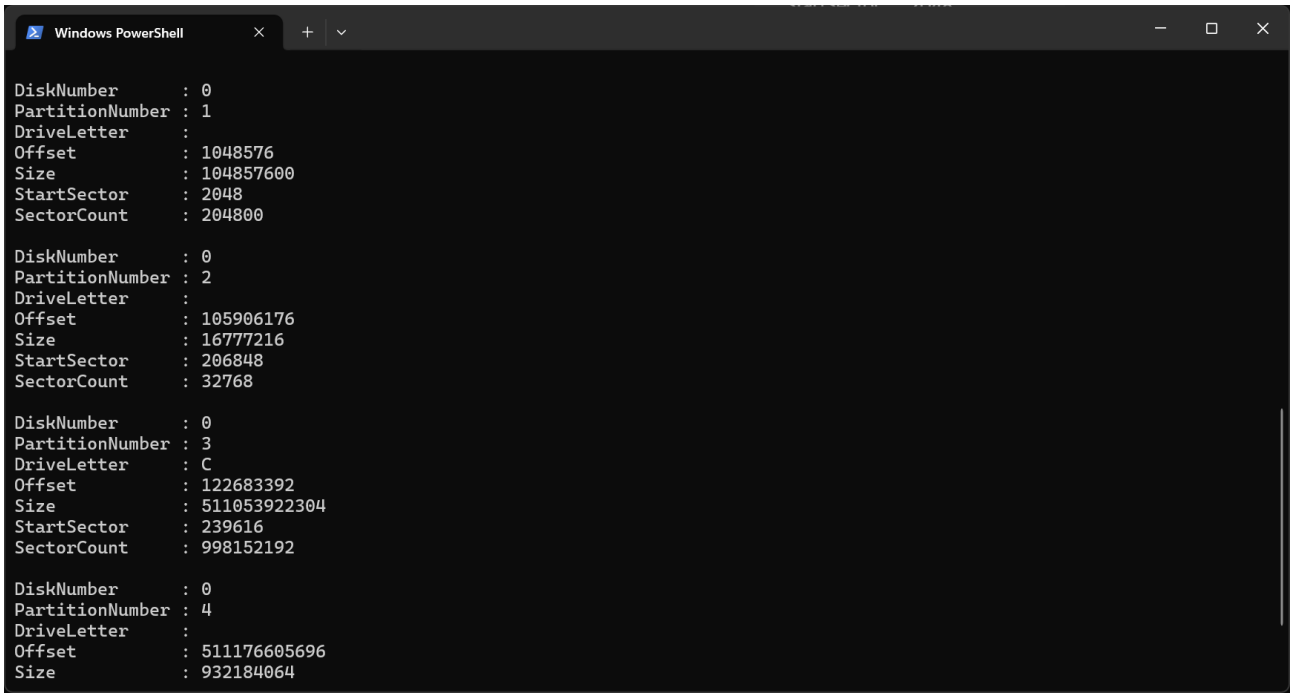
Part G

Task 1



Four Partitions with different sizes

Task 2/4



Partition 1
StartSector: 2048
SectorCount: 204800

Task 3
204800 x 512 / 1024 x 1024 x 1024 = 104,857,600 / 1,073,741,826 = 0.097GB

Task 5
Partition 2
StartSector: 206848
SectorCount: 32768

Task 6
32768 x 512 / 1024 x 1024 x 1024 = 16,777,216 / 1,073,741,826 = 0.015GB

Task 7

